Problem 6 | Problems T3_Diffusion.pdf #6 | Fick's 1st Law · Membrane Purification

Hydrogen Purification Through a Palladium Membrane

Original statement · Problems T3_Diffusion.pdf

6.- Determine the thickness of a plate of Pd *with transversal area of* $0.2\ \mathrm{m}^2$ *so that it can purify* $1.73 \times 10^{-3}\ \mathrm{kg/h}$ *of hydrogen, if the hydrogen concentration at the side of high pressure of the plate is* $1.5\ \mathrm{kg/m}^3$ *and at the side of low pressure is* $0.3\ \mathrm{kg/m}^3$*. The diffusion coefficient of hydrogen in* Pd *is* $1 \times 10^{-8}\ \mathrm{m}^2/\mathrm{s}$. Official Solution: Δx = 5 mm

PHASE 1 · HYPOTHESES & PARAMETERS

- **Pure Steady State:** Constant external pressures create time-invariant boundary concentrations C_{high} and C_{low} [Session 5 Slide 35].
- **Parameters:** $A = 0.2\ \mathrm{m}^2, \dot{m} = 1.73 \times 10^{-3}\ \mathrm{kg/h}\ C_1 = 1.5\ \mathrm{kg/m}^3, C_2 = 0.3\ \mathrm{kg/m}^3\ D = 1.0 \times 10^{-8}\ \mathrm{m}^2/\mathrm{s}$.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

1. Compute mass flux density [Session 5 Slide 15]:

$$J = \frac{\dot{m}}{A} \quad \left[\frac{\mathrm{kg}}{\mathrm{m}^2 \cdot \mathrm{s}}\right]$$

2. Apply Fick's First Law across the flat membrane [Slide 16, 35]:

$$J = D\frac{\Delta C}{\Delta x} \implies \Delta x = \frac{D \cdot (C_1 - C_2)}{J}$$

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

1. Flux Density (J):

$$\frac{\dot{m}}{A} = \frac{1.73 \times 10^{-3}\ \mathrm{kg/h}}{0.2\ \mathrm{m^2}} = 8.65 \times 10^{-3}\ \mathrm{kg/m^2 \cdot h}$$

$$J = \frac{8.65 \times 10^{-3}\ \mathrm{kg/m^2 \cdot h}}{3600\ \mathrm{s/h}} \approx \mathbf{2.4028 \times 10^{-6}}\ \mathrm{kg/m^2 \cdot s}$$

The worked solution stored in the text layer of Official Slide 35 gives $J = 8.65 \times 10^{-3}\ \mathrm{kg/m^2 \cdot h} = 2.4 \times 10^{-6}\ \mathrm{kg/m^2 \cdot s}$ *and* $\Delta x = 5\ \mathrm{mm}$ *(the answer box on the rendered slide appears empty).*

2. Concentration Difference:

$$\Delta C = C_1 - C_2 = 1.5\ \mathrm{kg/m^3} - 0.3\ \mathrm{kg/m^3} = 1.2\ \mathrm{kg/m^3}$$

3. Membrane Thickness (Δx):

$$\Delta x = \frac{(1.0 \times 10^{-8}\ \mathrm{m^2/s}) \times (1.2\ \mathrm{kg/m^3})}{2.4028 \times 10^{-6}\ \mathrm{kg/m^2 \cdot s}} \approx 4.994 \times 10^{-3}\ \mathrm{m} \approx \mathbf{5.0 \times 10^{-3}}\ \mathrm{m} = \mathbf{5}\ \mathrm{mm}$$

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

Dimensional Check: $[\Delta x] = \frac{(\mathrm{m^2/s}) \cdot (\mathrm{kg/m^3})}{\mathrm{kg/(m^2 \cdot s)}} = \mathrm{m}$. **Physical Insight:** At fixed concentrations the flux scales as $J \propto 1/\Delta x$ [Slide 16], so 5 mm is the thickness at which steady-state permeation equals the required purified flow. Hydrogen is small with respect to the matrix, so it diffuses interstitially, which is faster [Slide 13]; its diffusivity ($10^{-8}\ \mathrm{m^2/s}$) is far above the values of the diffusivity table [Slide 23]. This is the example of $\mathrm{H_2}$ purification through a Pd layer [Slide 35].